



## **Machine Learning II**

Data Science M.Sc.

Prof. Dr. Steffen Wagner  
Berliner Hochschule für Technik

Winter term 2025/26



# Linear and Quadratic discriminant analysis



## Bayes Classifier

- ▶ Suppose we have a classification problem with  $K$  possible classifications:  $y_1, y_2, \dots, y_K$ .
- ▶  $Y$  is our discrete or nominal outcome variable taking one of these values and depends on the predictor variables  $x_1, \dots, x_p$ .

A **Bayes Classifier** is a theoretical ideal. It assigns  $Y$  to the class  $y_j$  given the values of  $x_1, \dots, x_p$  according to the decision rule: Choose the group  $y_j$  which maximises the posterior probability

$$P(Y=y_k|x_1, \dots, x_p) \quad \text{for all } k=1, \dots, K$$

If  $K=2$  this simplifies to the rule, assign  $Y$  to  $y_2$  if

$$P(Y=y_2|x_1, \dots, x_p) > 0.5$$



## Linear discriminant analysis (LDA)

LDA is an algorithmic approach to replicating the Bayes classifier assuming the following model.

- ▶ We assume that the density of  $x$  within a given group  $k$  follows a normal distribution.
- ▶ The variance  $\sigma^2$  is assumed to be the same in all groups.

### Example:

We already considered classification on a small data set with 57 Students. For each student the height and sex is known.

- ▶ In the context to LDA the student height  $X$  is modelled depending on their sex.
  - ▶  $X|\text{male} \sim N(\mu_m, \sigma^2)$  and
  - ▶  $X|\text{female} \sim N(\mu_f, \sigma^2)$ .



## Example (cont)

- ▶ We have an *a priori probability* that a person chosen at random is male  $\pi_0 = P(\text{male}) = 0.5$  for the general population or  $\pi_0 = P(\text{male}) = 0.66$  if the population consists of engineering students.
- ▶ If we know a person's height, then we can do better than the prior probability: We can update the prior probability to give the posterior probability  $P(\text{male} | x = \text{height})$  by using *Bayes' Theorem*.
- ▶ Let  $\pi_0 = P(Y = \text{male})$  be the prior probability that a person chosen at random is male.
- ▶  $\pi_1 = P(Y = \text{male} | x)$  is the posterior probability that a person with height  $x$  is male.

$$\begin{aligned}\pi_1(x) := P(\text{male} | x) &= \frac{P(x | \text{male})P(\text{male})}{P(x | \text{male})P(\text{male}) + P(x | \text{female})P(\text{female})} \\ &= \frac{P(x | \text{male})\pi_0}{P(x | \text{male})\pi_0 + P(x | \text{female})(1 - \pi_0)}\end{aligned}$$



## Example (cont)

Because  $X|\text{male} \sim N(\mu_m, \sigma^2)$  we have a density function instead of a probability for

$$P(x|\text{male}) = f_{X|m}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x - \mu_m)^2}{2\sigma^2} \right\}$$

and  $X|\text{female}$  has a corresponding density. As we assume that the variance is the same in both classes.

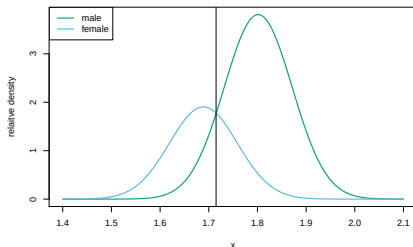
$$\begin{aligned} \pi_1(x) &= \frac{\pi_0 f_{X|m}(x)}{\pi_0 f_{X|m}(x) + (1 - \pi_0) f_{X|f}(x)} \\ &= \frac{\frac{\pi_0}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x - \mu_m)^2}{2\sigma^2} \right\}}{\frac{\pi_0}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x - \mu_m)^2}{2\sigma^2} \right\} + \frac{1 - \pi_0}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x - \mu_f)^2}{2\sigma^2} \right\}} \end{aligned}$$

- In practice we set  $\pi_0$  to be the proportion of males in the dataset and estimate  $\mu_m$ ,  $\mu_f$  via the sample mean in each class.
- $\sigma^2$  is the estimated variance of mean corrected  $x$ -values, either  $(x - \mu_m)$  or  $(x - \mu_f)$ .



## Example (cont)

- For  $x > 1.715$  we have  $\pi_1(x) > 0.5$  so the predicted value is “male”:

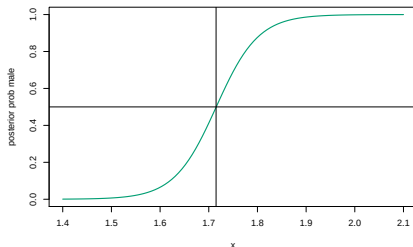


- This is in practical terms identical to the classifier developed using logistic regression. Logistic regression and LDA usually give similar results when there is only one predictor variable.
- The peak of the green line is higher, because in this data set two-thirds of the students were male, so  $\pi_0$  is set to  $\frac{2}{3}$



## Example (cont)

- ▶ The following curve shows the posterior probability of male  $\pi_1(x)$  as a function of height ( $x$ ):



- ▶ If the model is perfect, e.g. a normal distribution fits both groups and the mean for the male group  $\mu_m$  is exactly the sample mean  $\bar{x}_m$  etc., then the  $\pi_1(x) > 0.5$  corresponds to the Bayes classifier i.e. the best possible classifier.





## More than two classes

Linear discriminant analysis extends easily to  $K > 2$ .

- ▶  $P(Y=y_k|x)$  is the posterior probability our outcome variable is in group  $y_k$  given  $x$

$$P(Y=y_k|x) = \frac{f_{X|y_k}(x)P(Y=y_k)}{\sum_{j=1}^K f_{X|y_j}(x|Y=y_j)P(Y=y_j)}$$

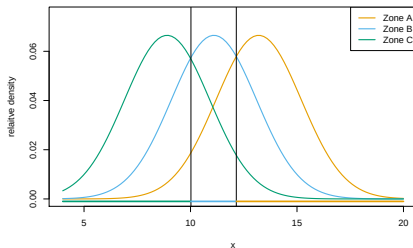
- ▶ Where  $P(Y=y_k)$  is the prior probability that our outcome variable is in group  $y_k$ .
- ▶ Our LDA classifier rule is to choose the value of  $k$  giving the largest  $P(Y=y_k|x)$



## Example: More than two classes

Fictional example:  $x$  = The basic rent per square meter for new contracts for flats rented in and around Berlin in 2017:

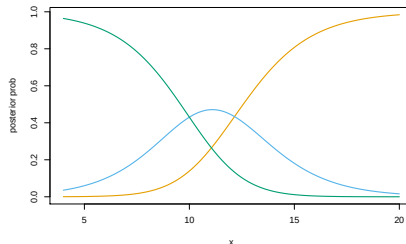
- $Y$  = BVG Zone A, B, or C. The prior probabilities are  $\frac{1}{3}$  for each zone:





## Example: cont.

The Posterior probability  $P(Y=y_k|x)$  of being in each zone<sup>1</sup> given the rent  $x$  is plotted and the classifier rule is choose the curve with the highest density for a given value of  $X$ :



The classification rule results to:

Zone A    when                     $x > 12.15\text{€}$

Zone B    when  $10.02\text{€} < x < 12.15\text{€}$

Zone C    when                     $x < 10.02\text{€}$

---

<sup>1</sup> same colour coding as on the previous slide



## More predictor variables

Suppose we have 2 or more predictor variables,  $p \geq 2$ .

The distribution in each group is now modelled using a multivariate normal distribution.

$$(X_1, X_2, \dots, X_p) | Y=y_k \sim N(\mu_k, \Sigma^2)$$

with  $\mu_k$  is a vector of length  $p$ .  $\Sigma^2$  is the common variance-covariance matrix.



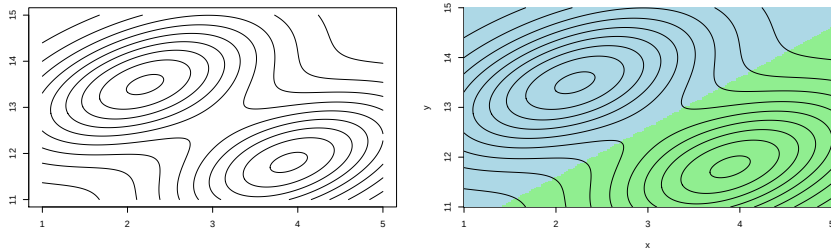
## More predictor variables (cont)

- ▶ There are  $p=2$  variables  $K=2$  and two classes for  $Y$ .
- ▶ In the group  $Y=1$ , the mean of  $x_1$  is 2.2, and the mean of  $x_2$  is 13.5, so  $\mu_1 = \begin{pmatrix} 2.2 \\ 13.5 \end{pmatrix}$
- ▶ In the group  $Y=2$ , the mean vector is  $\mu_2 = \begin{pmatrix} 3.9 \\ 11.8 \end{pmatrix}$ .
- ▶ The variance matrix for all observations is  $\Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$ .
- ▶ The prior probability that  $Y=1$  is  $\pi_0 = 0.5$ .



## More predictor variables (cont)

The equivalent to the plot on slide 7, when there are two predictor variables, looks as follows:



- ▶ For a given value of  $(x_1, x_2)$ , we can calculate the density of  $(x_1, x_2)$  given  $Y=1$  (blue) and the density of  $(x_1, x_2)$  given  $Y=2$  (green) and plot these as a contour plot.
- ▶ The blue region is where the blue contours are higher than the green contours.
- ▶ The green region is where the blue contours are lower than the green contours.



## More predictor variables (cont)

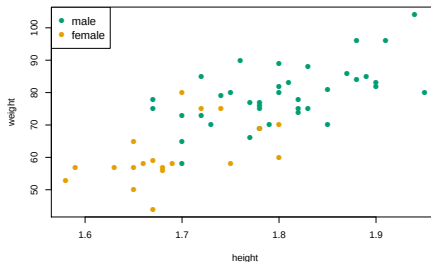
- ▶ The posterior probabilities are  $P(Y=1|x_1, x_2)$  and  $P(Y=2|x_1, x_2)$ .
- ▶ The blue region corresponds exactly to the area where  $P(Y=1|x_1, x_2)$  is greater than  $P(Y=2|x_1, x_2)$ .
- ▶ The green region corresponds exactly to the area where  $P(Y=1|x_1, x_2)$  is less than  $P(Y=2|x_1, x_2)$ .
- ▶ The boundary between the two is linear. This is why it is called linear discriminant analysis.



## Example: student sex based on height and weight

LDA on the student data using height and weight.

Scatter plot: original data





The LDA algorithm gives the following coefficients for height and weight

The prediction boundary is determined by whether this score is greater than a cut off value (here -13.6), which is shown in the scatter plot:





## Quadratic discriminant analysis (QDA)

- ▶ In LDA we assumed that the covariance matrix is the same in each group.
- ▶ If we relax this assumption, allowing for a different covariance in each group, then we have **quadratic discriminant analysis**.
- ▶ Each class is still modelled using a using a multinomial distribution.

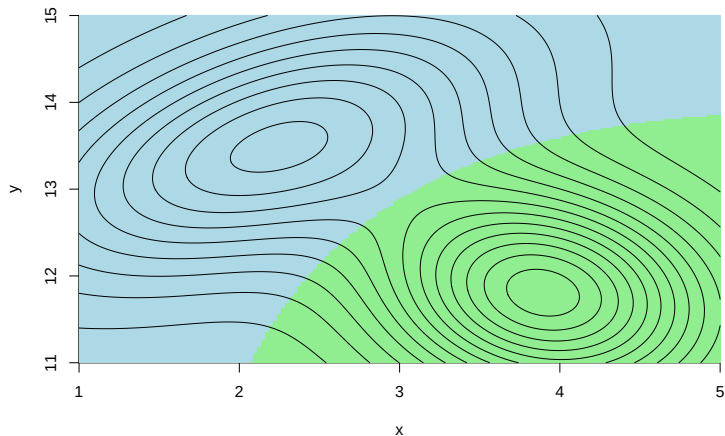
$$(X_1, X_2, \dots, X_p) | Y=y_k \sim N(\mu_k, \Sigma_k^2)$$

$\mu_k$  is a vector of length  $p$  for class  $k$  and  $\Sigma_k^2$  is the  $p \times p$  variance-covariance matrix for class  $k$ .



## QDA (cont)

The boundary between the two regions is now a curve.





## Example: student data

### Comparison of results using the student data:

LDA: only variable height  
(same as with logistic regression)

|            | Classified as |      |
|------------|---------------|------|
|            | Female        | Male |
| Actual     | 0             | 1    |
| Female (0) | 13            | 6    |
| Male (1)   | 5             | 33   |

⇒ Sensitivity = 0.868

Specificity = 0.684.

LDA: using height and weight  
is noticeably better.

|            | Classified as |      |
|------------|---------------|------|
|            | Female        | Male |
| Actual     | 0             | 1    |
| Female (0) | 14            | 5    |
| Male (1)   | 2             | 36   |

⇒ Sensitivity = 0.947

Specificity = 0.736.

QDA gives no further improvement over LDA .